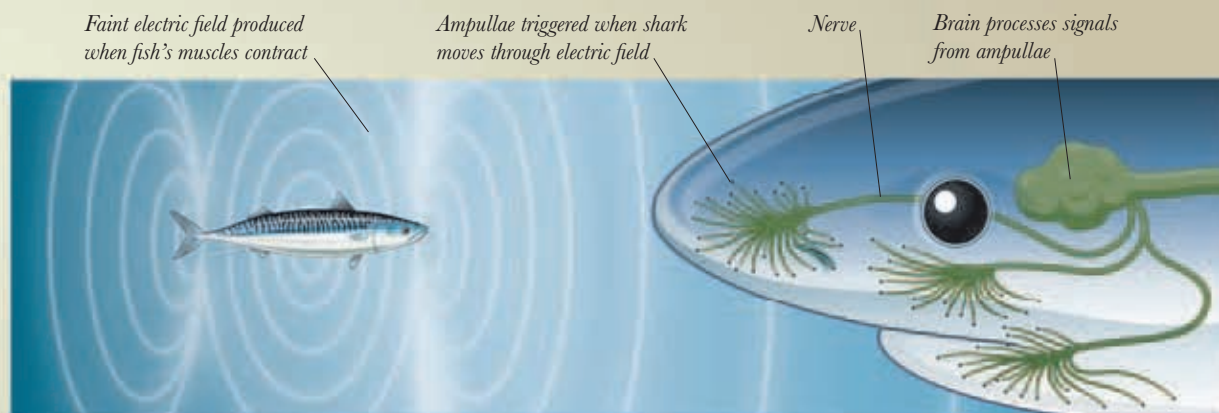


Electroreception

Some fish can produce massive electric shocks, which they use to stun or kill their prey. Many more fish, such as sharks and rays, sense the electric fields created by other fish. A shark does this with gel-filled sensors, or ampullae, clustered in its head. It can detect differences of a millionth of a volt, guiding the shark toward the source of the field.

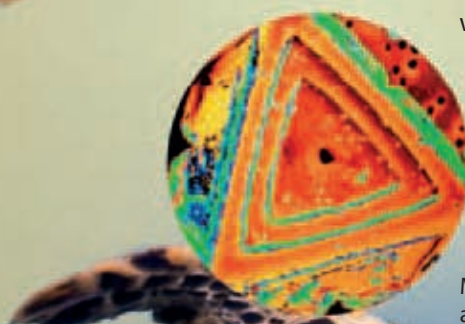


Magnetic compass

When green turtles breed, they usually head back to the beaches where they were born, sometimes traveling vast distances.

Like migrating birds, the turtles navigate in several ways. In addition to steering by looking at the Sun and stars, they may be guided by crystals of magnetite—an iron-based mineral in their brains. Magnetite is the most magnetic natural mineral on Earth and is found in many animals that steer by Earth's magnetic field.

Magnified view of a magnetite crystal



Feeling pressured

Animals cannot forecast the weather, but many can sense changes in air pressure. A gradual rise usually means the weather is becoming drier and more settled, while a sudden drop means stormy weather may be on its way. Swifts often travel away from low pressure air, returning when the pressure rises again.



Magnetic architects

In Australia, magnetic termites build slab-shaped nests with narrow edges pointing north and south. As the Sun travels east to west, there is always one face in shadow where termites can keep cool, and at noon, only the top edge is exposed to direct sunlight. Worker termites are blind, so they cannot use the Sun as a guide to line up their nests. Instead, they sense Earth's magnetic field.